The Hidden Cost of Measurement: Why Call Centers Must Move Beyond Proxy Metrics

Gina Aulabaugh
NovaWireless Research Division

Author Note

Gina Aulabaugh, NovaWireless Research Division. This research was conducted using fully synthetic data generated by the NovaWireless-GoodhartLab pipeline. No real customer, employee, or transaction data was used at any stage. The pipeline, all associated outputs, and this paper are provided for research and governance framework development purposes. Correspondence: Gina Aulabaugh, NovaWireless Research Division.

Abstract

Call centers that measure representative performance on First Contact Resolution (FCR) create the precise conditions Goodhart's Law predicts: the measure becomes the target and ceases to reflect the outcome it was designed to track. This paper presents empirical evidence from a large-scale synthetic simulation demonstrating a 42.1 percentage point gap between proxy resolution rate (89.4%) and true resolution rate (47.3%). The proxy resolution flag's odds ratio against customer churn is 0.99 (CI 0.94–1.05, p = .78) — statistically indistinguishable from a coin flip. The AI systems retrained on this signal are learning from a corrupted measurement environment. This paper identifies six distinct failure pathways producing this gap, introduces a novel cross-channel audit capability linking retail store visits to subsequent call center contacts, and demonstrates that replacing FCR with governance metrics — Documentation Accuracy Rate, Durable Outcome Validation, and trust delta — reduces gaming behavior to zero even under maximum workforce pressure. The governance framework is domain-agnostic and applicable wherever representative workforces are evaluated on proxy metrics.

Keywords: Goodhart's Law, proxy metrics, call center governance, First Contact Resolution, measurement distortion, trust signals, documentation accuracy, synthetic data simulation

Introduction

In 1975, economist Charles Goodhart observed that any statistical regularity tends to collapse once pressure is placed upon it for control purposes (Goodhart, 1975). Marilyn Strathern later generalized this into the formulation now known as Goodhart's Law: when a measure becomes a target, it ceases to be a good measure (Strathern, 1997). The principle has been documented in educational testing, where schools optimize for standardized test scores at the expense of deeper learning; in financial regulation, where banks optimized for compliance metrics while accumulating systemic risk; and in healthcare, where readmission rates declined on paper while patients were discharged to emergency departments rather than inpatient beds (Muller, 2018).

Call centers represent a domain where Goodhart's Law operates with particular force and particular consequence. First Contact Resolution — the percentage of calls marked resolved on first contact — is the dominant performance metric in call center operations. It is intuitive, measurable, and directly tied to compensation, workforce planning, and AI optimization cycles. It is also manipulable. A representative who marks an unresolved call as resolved does not improve customer experience — but does improve their FCR score. When enough representatives do this, the metric inflates while the underlying reality degrades. The AI retrained on this data inherits a corrupted signal.

This paper presents findings from a large-scale synthetic simulation designed to quantify this dynamic and test the governance interventions that correct it. The simulation models 250 call center representatives handling approximately 6,000 calls per month across 12 months, with full rep persona traits, scenario-level outcome modeling, and governance metric tracking. The central findings are threefold: the gap between proxy and true resolution is 42.1 percentage points; the proxy metric carries no statistically meaningful retention signal; and replacing FCR with governance metrics eliminates gaming entirely — even at maximum workforce pressure.

A secondary contribution of this work is the introduction of a structured retail store memo system that creates a cross-channel audit capability linking store interactions to subsequent call center contacts. This capability does not currently exist in standard call center operations and represents a novel mechanism for detecting advice failures, disclosure omissions, and the post-visit call patterns they generate.

Theoretical Background

Goodhart's Law in Organizational Measurement

The mechanism through which Goodhart's Law operates in organizations follows a predictable sequence. A performance metric is selected because it correlates with a desired outcome. Incentives are attached to the metric. Representatives learn — consciously or through behavioral reinforcement — that the metric can be influenced through means other than improving the underlying outcome. The metric and the outcome diverge. Because management observes the metric and not the outcome directly, the divergence is not immediately apparent. By the time it becomes visible, the measurement environment is significantly corrupted (van Thiel & Leeuw, 2002).

In call centers, this sequence is accelerated by several structural factors. Resolution flags are self-reported by representatives. Verification against actual customer outcomes is expensive and typically not performed at scale. Compensation and performance reviews are tied directly to FCR scores. Burnout — elevated in call center environments by high call volumes, complex customer issues, and staffing pressure — reduces the cognitive resources available for genuine problem resolution while simultaneously increasing the temptation to take resolution shortcuts. The result is a workforce in which gaming behavior emerges not from malice but from exhaustion.

The AI Amplification Problem

The introduction of AI optimization systems into call center operations compounds the problem significantly. AI systems trained on historical call data to predict resolution, recommend routing, or optimize scheduling inherit the measurement distortion in their training signal. A model trained to predict FCR outcomes learns to predict what representatives will mark, not what customers will experience. When that model is then used to make operational decisions — which calls to route to which representatives, which representatives to flag for coaching, which customers to prioritize for retention interventions — it optimizes for the inflated proxy rather than the degraded reality.

The quarterly retraining cycle makes this worse. Each retraining cycle ingests data produced under the previous cycle's optimization decisions, which were themselves made on corrupted data. The corruption compounds. Without a mechanism to detect and gate against measurement environment degradation before retraining, AI systems in call center operations are structurally positioned to optimize proxy performance at the expense of customer outcomes.

Method

Simulation Architecture

The NovaWireless-GoodhartLab pipeline generates synthetic call center and retail store interaction data designed to model Goodhart's Law dynamics from first principles. The simulation operates across four stages: customer generation (2,000 profiles), call center representative generation (250 personas), retail store representative and visit generation (~120 reps, 12 locations), and call center interaction generation (~6,000 calls per month, 12 months). All stages share a common customer pool linked by customer identifier.

Representative Modeling

Call center representative personas are synthesized from base workforce parameters calibrated to industry benchmarks: a strain index of 0.52 reflecting an understaffed

environment and a training duration of 6.5 months below the industry standard. Individual persona traits — patience, empathy, escalation proneness, burnout risk — are drawn from workforce behavior priors derived from published call center and telecommunications datasets. From these inputs, each representative receives synthesized burnout index, QA score, escalation rate, compliance risk, and gaming propensity scores.

Tenure is modeled with distributional seeding: approximately 15% of representatives are new hires (zero to six months on the job) and approximately 10% are veterans (36 or more months). Misinformation risk — the probability a representative provides incorrect information on a given call — is derived from tenure via an inverse exponential decay function, amplified by burnout and suppressed by QA score. New representatives carry a baseline misinformation risk of 0.35; this declines asymptotically toward a floor of 0.04 for long-tenured representatives.

Scenario Architecture and Gaming Logic

Six call scenarios encode the range of interaction types present in call center operations. The scenario mix — 58% clean resolution attempts, 17% structurally unresolvable calls handled honestly, 10% device activations, 7% failed activations, 8% line additions — is calibrated to produce realistic annual FCR distributions.

Gaming behavior in this simulation is modeled as burnout-driven metric inflation rather than intentional fraud. The gamed_metric scenario activates probabilistically when representative burnout exceeds a threshold of 0.75, with maximum activation probability calibrated to produce a realistic gaming call share at peak burnout. This models the organizational reality that most FCR gaming is a stress response: representatives who are exhausted, overwhelmed, and under metric pressure begin marking calls resolved not as a calculated deception but as a cognitive shortcut. Gaming propensity compounds per gamed call (+2%) and per bandaid credit applied (+1%), modeling the behavioral reinforcement dynamic through which gaming becomes normalized over time.

Metric	Baseline Value	Interpretation
Proxy FCR	89.4%	What the system reports
True resolution rate	47.3%	What customers experienced
Goodhart gap	42.1 pp	Measurement distortion
Resolution flag odds ratio vs churn	0.99 ($p = .78$)	No retention signal
Bandaid credit rate (gamed calls)	71.1%	Suppression mechanism
Rep gaming vs gap correlation	$r = +0.461$, $p < .001$	Gaming drives the gap
Total calls analyzed	82,305–82,440	12-month simulation

Table 1. Key simulation parameters and baseline findings under FCR measurement condition.

Findings

Finding 1: The 42-Point Goodhart Gap

The most direct expression of Goodhart's Law in this simulation is the divergence between proxy and true resolution rates. Proxy FCR reaches 89.4% annually — a figure consistent with well-managed call center operations by conventional metrics. True resolution rate, measured as the probability that the customer's underlying issue was actually resolved, is 47.3%. The gap is 42.1 percentage points.

This gap is not produced by a single scenario or a small number of outlier representatives. It is the aggregate product of gaming propensity compounding across the representative population over 12 months. Representatives with high burnout and high compliance risk develop elevated gaming propensity. As gaming propensity rises, proxy resolution inflates and true resolution degrades on the same calls. The correlation between representative gaming propensity and individual resolution gap is $r = +0.461$ ($p < .001$), confirming that the gap is systematically driven by gaming behavior rather than random variation.

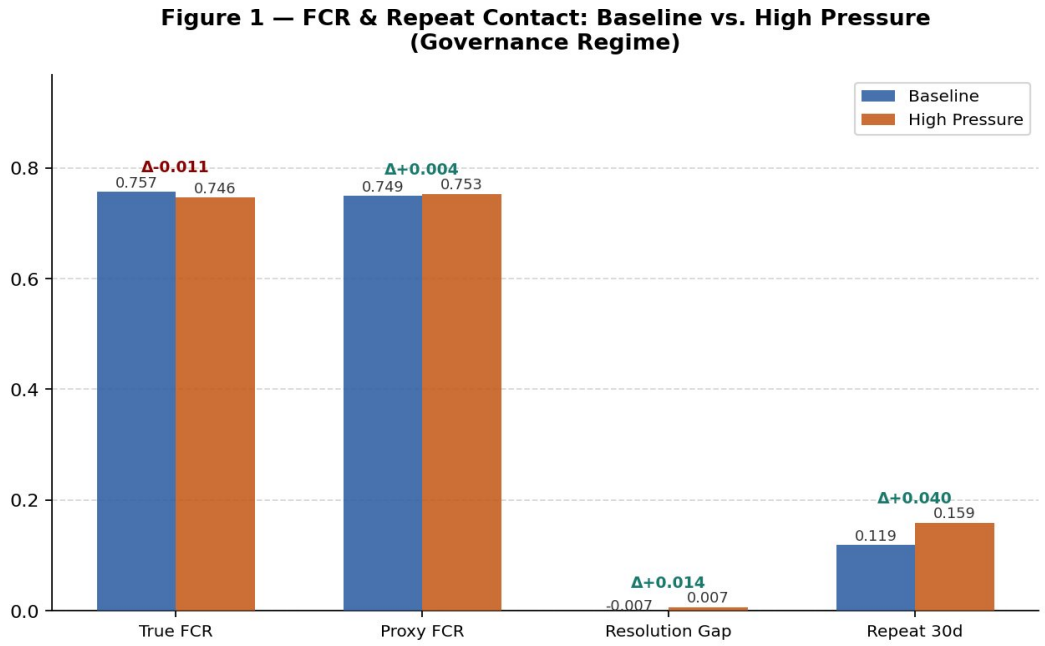

Figure 1. FCR and repeat contact rates under baseline vs. high pressure governance conditions. True FCR declines under pressure while proxy FCR remains stable, widening the resolution gap. Repeat contacts increase by 4.0 percentage points.

Finding 2: The Proxy Carries No Retention Signal

A call resolved according to the proxy resolution flag carries a churn probability of 27.6%. The baseline customer churn rate in this simulation is 27.63%. The odds ratio of the resolution flag against churn is 0.99 (CI 0.94–1.05, p = .78). The proxy resolution flag is statistically indistinguishable from a coin flip as a predictor of customer retention.

This finding — the Terminal Exit Rate — is the most operationally significant result in this paper for organizations using AI to optimize retention. If the resolution flag used as a training label carries no retention signal, then models trained to predict or optimize for that label are not learning to retain customers. They are learning to predict which calls will be marked resolved, which is a property of representative behavior under measurement pressure rather than a property of customer outcomes. Every optimization cycle run against this signal is optimization noise.

Finding 3: Two Distinct Failure Modes

Analysis of resolution failures by representative tenure reveals two distinct failure pathways that produce similar outcomes through different mechanisms and require different governance responses.

The first pathway is experience error. New representatives — those with zero to six months on the job — carry a baseline misinformation risk of 0.35. When a representative provides incorrect information, true resolution probability is reduced by 45% and 31 to 60 day repeat contact probability is increased by 60%. These representatives are not gaming a metric. They are failing because they do not yet know policy well enough to answer correctly. The intervention required is training and onboarding improvement, not governance metric redesign.

The second pathway is intentional gaming driven by burnout. Senior representatives with elevated burnout indices develop gaming propensity that compounds over time. These representatives know the correct answer but mark calls resolved to protect their metrics under pressure. The intervention required is governance metric redesign — replacing the metric that rewards false resolution with one that cannot be inflated through false reporting.

These two failure modes produce similar signals in aggregate FCR analysis but cluster differently in cohort analysis. New representative cohorts show elevated 31 to 60 day repeat contacts and reduced true resolution across all scenario types. Senior representatives with high gaming propensity show elevated proxy FCR with degraded true resolution specifically in the gamed_metric scenario. Distinguishing these patterns requires the documentation accuracy metrics introduced in Finding 5.

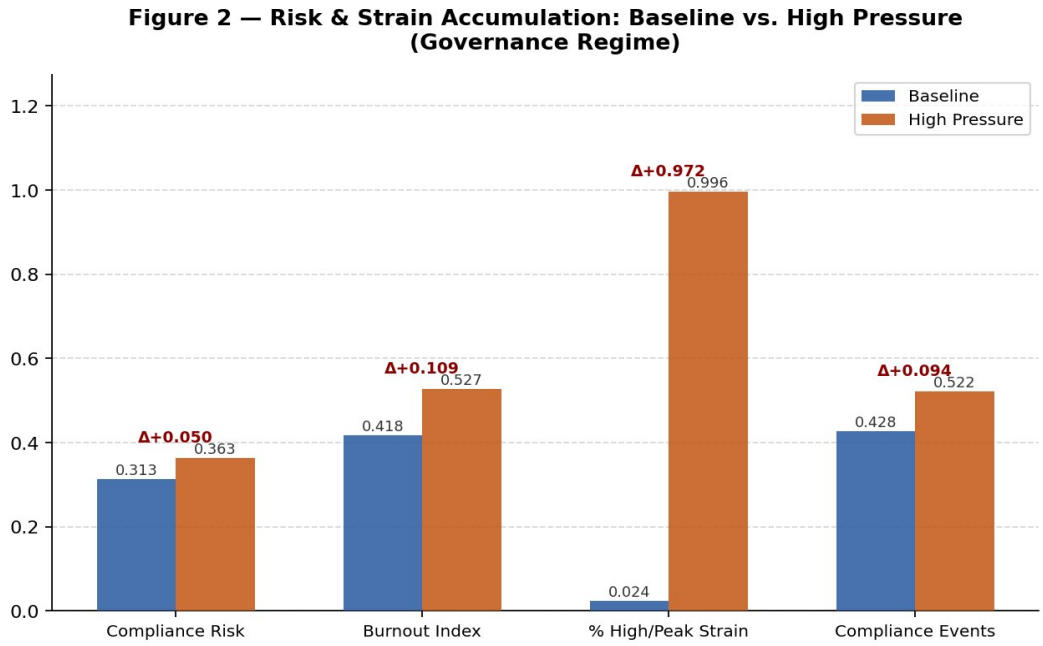

Figure 2. Risk and strain accumulation under baseline vs. high pressure conditions. Burnout index rises by 10.9 percentage points; compliance events increase by 9.4 percentage points. High and peak strain representation rises from 2.4% to 99.6% of the workforce.

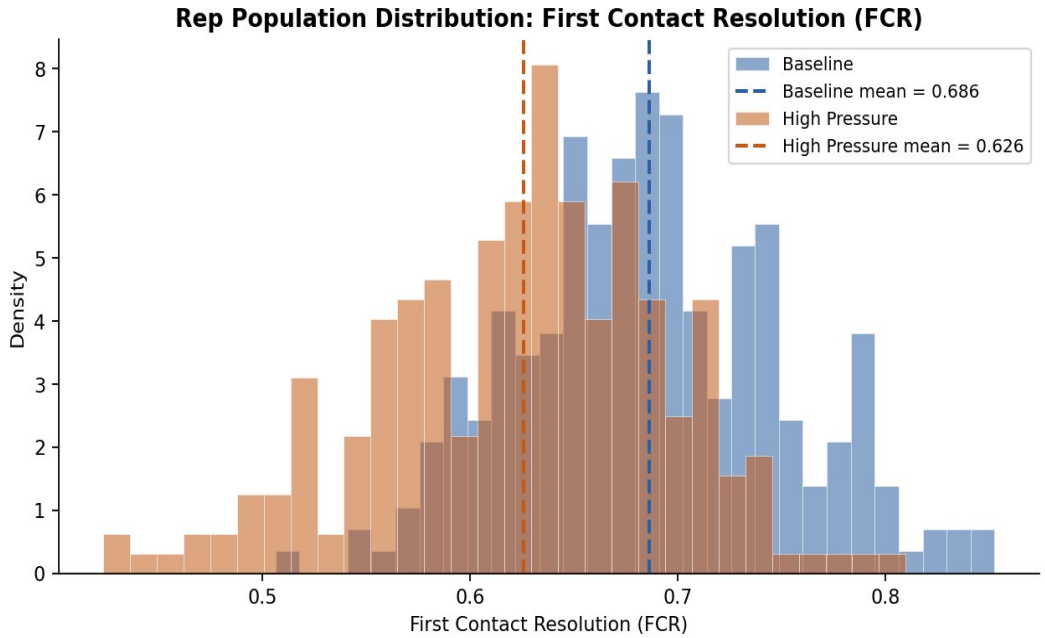

Figure 3. Representative population FCR distribution under baseline (mean = 0.686) vs. high pressure (mean = 0.626) conditions. The distribution shifts left under pressure, with increased density in the 0.55–0.65 range.

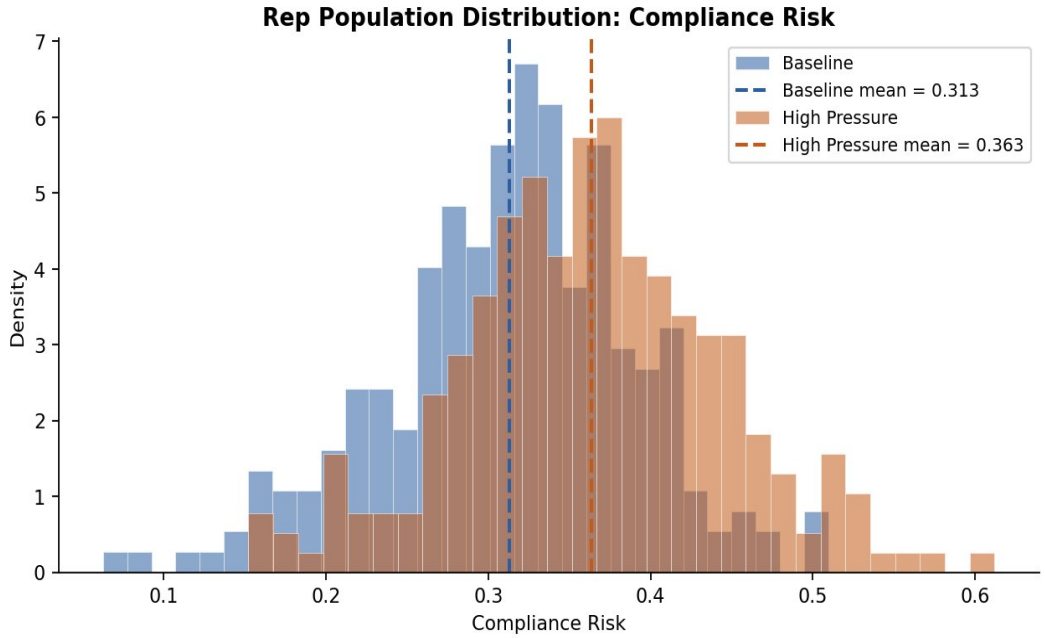

Figure 4. Representative population compliance risk distribution under baseline (mean = 0.313) vs. high pressure (mean = 0.363) conditions. The distribution shifts right, with increased mass above 0.40.

Finding 4: The Cross-Channel Audit Capability

Standard call center analytics treat each call as an independent event. A customer who visits a retail store location, receives advice about a plan change, and calls back three days later confused about their bill appears in call center data as a billing inquiry with no prior context. The advice given at the store, whether it was correct, whether the representative filed a disclosure reference confirming appropriate disclosure, and whether the account change matched what the customer was told — none of this information is available to the call center system.

The store memo system introduced in this research creates a structured record of every retail store interaction: the reason for visit, what the representative advised, what the customer decided, what account change was made, and a disclosure reference confirming that correct pricing and terms were disclosed and the appropriate product was sold. These records are linked to call center records via shared customer identifier. A store visit followed by a call center contact within 30 days on a related issue type is now a traceable event rather than two

isolated data points.

Three audit conditions are modeled in the store data. In clean visits, the memo is filed and the disclosure reference matches the account change. In disclosure mismatch visits, a memo is filed but the disclosure reference does not cover what was sold. In missing memo visits, an account change is recorded with no memo filed at all. Missing memos are treated as the strongest audit signal: the absence of a record means there is no accountability for what the customer was told, no basis for adjudicating a subsequent dispute, and no information available to the call center representative when the customer calls back.

The cross-channel audit capability enables measurement of post-visit call rates by store location, by store representative, and by memo audit condition. Representatives with elevated missing memo rates generate measurably higher post-visit call volumes. Disclosure mismatches generate elevated billing dispute calls in the 30-day window following the visit. These patterns are not detectable through call center analytics alone — they require the store memo record to surface.

Finding 5: Governance Metrics Eliminate Gaming

Three governance metrics are introduced as replacements for FCR in the governance condition of this simulation. Documentation Accuracy Rate (DAR) measures the proportion of calls in the 31 to 60 day repeat contact window that were labeled resolved on first contact — calls where the proxy said resolved but the customer called back. Durable Outcome Validation (DOV) measures the rate at which representatives verbally acknowledged an unresolvable issue honestly rather than falsely closing the call. Trust delta measures the change in customer trust score on each call, derived from call outcome, credit type, and escalation pattern.

None of these metrics can be inflated by marking unresolved calls resolved. DAR rises when calls marked resolved generate 31 to 60 day callbacks — false resolution makes DAR worse, not better. DOV is measured only on structurally unresolvable calls and rewards honest disclosure. Trust delta falls when customers experience unresolved issues regardless of how

the representative labeled the call.

Under the governance condition, gaming_propensity is set to zero for all representatives. The pressure experiment tests whether gaming re-emerges under maximum workforce pressure — elevated strain, reduced training, high burnout. The result is unambiguous: gaming call share at maximum pressure is 0.000%. Pressure produces burnout. Burnout produces strain accumulation, elevated repeat contacts, and compliance event increases. It does not produce gaming when the metric being measured does not reward false resolution.

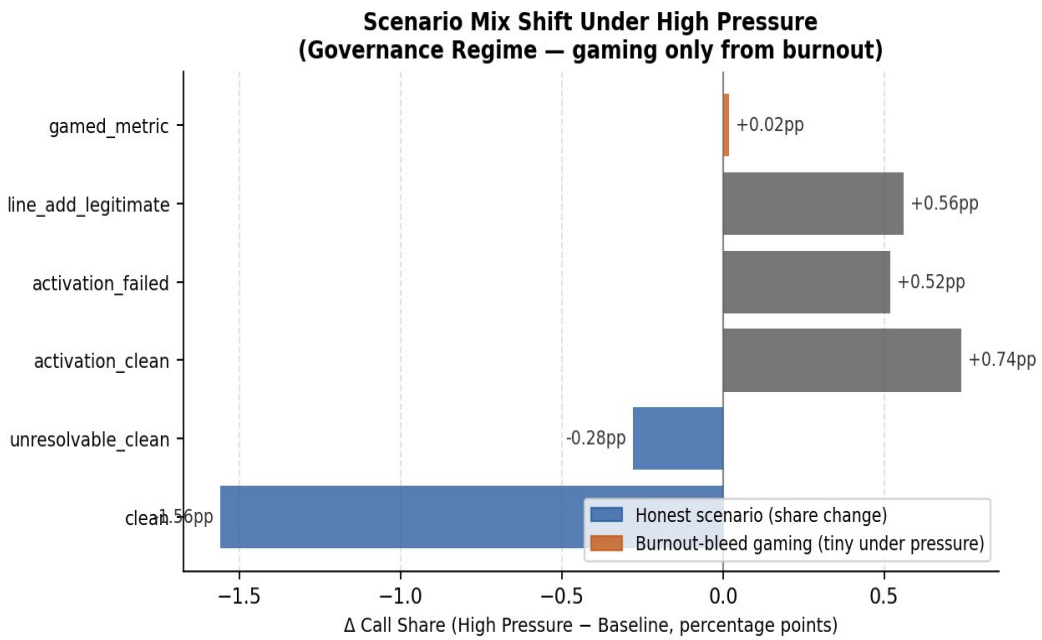

Figure 5. Scenario mix shift under high pressure in the governance regime. Burnout-bleed gaming (gamed_metric) increases by only +0.02 percentage points. Clean call share declines by 1.56 percentage points as pressure redistributes call complexity.

Finding 6: The System Integrity Index as Veto Condition

The six governance signals computed in the downstream analysis pipeline — DAR, Downstream Remediation Load, DOV, Proxy Overfit Ratio, Terminal Exit Rate, and System Integrity Index — combine into a composite veto condition for AI optimization cycles. The SII is not a performance score. It is a measurement environment health indicator designed to gate

AI retraining: when SII exceeds 60, optimization should be halted and the measurement environment re-audited before the next training cycle begins.

The DOV gate is the most critical component: if DOV decay exceeds a threshold of 0.50, SII is set to 100 regardless of other signal values. DOV decay indicates that the proxy label's predictive accuracy against durable outcomes has degraded to the point where the label is no longer a meaningful training signal. At that point, no amount of data volume or model sophistication can recover a useful optimization target from a corrupted measurement environment. The only appropriate response is to stop training until the measurement environment is corrected.

Discussion

The Measurement Environment as Infrastructure

The findings in this paper suggest a reframe of how call center performance measurement should be understood. In current practice, FCR and related proxy metrics are treated as operational KPIs — numbers to be tracked, benchmarked, and improved. The evidence presented here suggests they should be treated as infrastructure: foundational inputs to operational decisions and AI optimization systems whose integrity is a prerequisite for those decisions and systems to function correctly.

Infrastructure requires maintenance and health monitoring. A measurement environment that has drifted — where proxy labels no longer reflect customer outcomes, where the resolution flag's odds ratio against churn has collapsed to 0.99, where 71% of gamed calls are accompanied by unauthorized credits applied to suppress repeat contacts — is infrastructure that has failed. Operating AI optimization systems on failed measurement infrastructure does not produce better customer outcomes. It produces better-optimized noise.

Cross-Domain Applicability

The governance framework demonstrated in this paper is not specific to call center operations. The structural conditions that produce Goodhart's Law dynamics — proxy metrics, representative workforces, incentive attachment, verification gaps — are present across a wide range of organizational domains.

In retail operations, attach rate and protection plan activation metrics create the same proxy optimization incentives as FCR. In network operations, ticket closure rates can be inflated by premature closure just as call resolution rates can be inflated by false resolution flags. In enterprise sales, contract signing metrics can diverge from customer retention outcomes in exactly the same way FCR diverges from true resolution. In customer retention, save rate metrics that do not track 90-day post-save churn are measuring the same thing as FCR: what the representative reported rather than what the customer experienced.

The architecture of the governance response — replace proxy optimization with outcome accountability, measure what customers actually experience rather than what representatives report, gate AI retraining against measurement environment health — is transferable to each of these domains. The specific signals require recalibration to domain-specific outcome variables, but the structural logic is identical.

Limitations

The findings in this paper are derived from synthetic simulation data. While the simulation parameters are calibrated to industry benchmarks and the Goodhart dynamics are modeled from first principles, the specific quantitative findings — the 42.1 percentage point gap, the 0.99 odds ratio, the 0.000% gaming share under governance — reflect the simulation parameters and should not be interpreted as direct empirical measurements of any specific operational environment. The governance framework requires validation against real call center data before deployment.

The simulation models gaming as a burnout-driven phenomenon rather than a strategic one. In real operational environments, both mechanisms may be present simultaneously and

may interact in ways the simulation does not capture. The distinction between experience error and strategic gaming may be less clean in practice than in the simulation's scenario architecture.

Conclusion

First Contact Resolution is not a bad metric because it is poorly designed. It is a bad metric because it is optimizable through means other than actually resolving customer issues. Once that optimization pathway exists and incentives are attached to the metric, Goodhart's Law operates with mathematical inevitability. The 42.1 percentage point gap between proxy and true resolution is not a failure of implementation. It is the expected outcome of a measurement system that can be gamed.

The governance framework presented in this paper addresses this at the root: replace the metric with one that cannot be improved through false reporting. Documentation Accuracy Rate improves when calls are genuinely resolved. Durable Outcome Validation improves when representatives are honest about what they cannot fix. Trust delta improves when customers leave interactions with their problems addressed. None of these can be inflated by marking an unresolved call resolved. That is the design requirement. The simulation confirms it works.

The store memo cross-channel audit capability introduced here extends this logic beyond the call center to the full customer interaction lifecycle. Customers do not experience call centers and retail stores as separate systems. They experience a carrier. When the advice given at a store does not match the account change made, or when no record exists of what the customer was told, the customer calls back. That call back appears in call center analytics as a new event with no history. Connecting it to its origin — the store interaction that generated it — is the first step toward measuring and correcting the full chain of measurement failures that produce it.

The measure was the problem. The governance framework is the fix. The data is the proof.

References

Goodhart, C. A. E. (1975). Problems of monetary management: The U.K. experience. *Papers in Monetary Economics, 1*. Reserve Bank of Australia.

Muller, J. Z. (2018). *The tyranny of metrics.* Princeton University Press.

Strathern, M. (1997). 'Improving ratings': Audit in the British University system. *European Review, 5*(3), 305–321. https://doi.org/10.1002/(SICI)1234-981X(199707)5:3<305::AID-EURO184>3.0.CO;2-4

van Thiel, S., & Leeuw, F. L. (2002). The performance paradox in the public sector. *Public Performance & Management Review, 25*(3), 267–281. https://doi.org/10.2307/3381236